

SURDS & INDICES

Law of Indices: $a \times a \times a = a^3$ power
Base

$$a^m \times a^n = a^{m+n} \quad (a \neq 0)$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{m \cdot n}$$

$$a^m = a^n \rightarrow m = n$$

$$a^m = b^m \rightarrow a = b$$

Surds

$$2^5 = 32 \rightarrow 2 = 32^{\frac{1}{5}}$$

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

order of surds

$$\sqrt[5]{32}$$

radicand

Law of surds:-

$$\sqrt[n]{a} = a^{\frac{1}{n}}$$

$$\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b} = a^{\frac{1}{n}} \times b^{\frac{1}{n}} = (ab)^{\frac{1}{n}}$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \left(\frac{a}{b}\right)^{\frac{1}{n}}$$

$$(\sqrt[n]{a})^m = \sqrt[n]{a^m} = a^{m \times \frac{1}{n}}$$

$$\sqrt[n]{m \sqrt[n]{a}} = \sqrt[n]{\frac{1}{a^{\frac{1}{m}}}} = a^{\frac{1}{mn}} = \sqrt[n]{a^{\frac{1}{m}}}$$

$$(\sqrt[n]{\sqrt[n]{a}})^m = a^{\frac{m}{n^2}}$$

$$(3+\sqrt{5})^2 = \sqrt[2]{4+6\sqrt{5}}$$

$$\sqrt{a^2+b^2-2ab} \begin{cases} a-b & (a > b) \\ b-a & (b > a) \end{cases}$$

$$(a-b)^2 = (b-a)^2 = a^2+b^2-2ab$$

Ex 1

$$\sqrt{2} = 1.414$$

$$\sqrt{3} = 1.732$$

$$\sqrt{5} = 2.24$$

$$\sqrt{6} = 2.45$$

$$\sqrt{10} = 3.16$$

Approx. root:-

$$\sqrt{\frac{a+b}{2}}$$

$$\sqrt{17} \approx 13 + \frac{8}{2 \times 13} \approx 13.309$$

RATIONALISATION

$$\frac{1}{3+2\sqrt{2}} = \frac{1}{3+2\sqrt{2}} \times \frac{3-2\sqrt{2}}{3-2\sqrt{2}} = \frac{3-2\sqrt{2}}{3^2-(2\sqrt{2})^2}$$

$$\frac{3-2\sqrt{2}}{9-8} = \frac{3-2\sqrt{2}}{1} = 3-2\sqrt{2}$$

Ex 2

$$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ac$$

$$= a^2+b^2+c^2+2[ab+bc+ac]$$

$$(a-b+c)^2 = a^2+b^2+c^2+2ab-2bc+2ac$$

$$= a^2+b^2+c^2+2[ab-bc+ac]$$

$$(a+b-c)^2 = a^2+b^2+c^2+2ab-2bc-2ac$$

$$= a^2+b^2+c^2+2[ab-bc-ac]$$

$$n^2 = (\sqrt{a+b} \pm \sqrt{a-b})^2$$

$$a^2-ab+b^2 = \frac{a^3+b^3}{a+b}$$

$$a^2+ab+b^2 = \frac{a^3-b^3}{a-b}$$

Infinite series

$$\sqrt{a \cdot \sqrt{a \cdot \sqrt{a \dots \infty}}} = \textcircled{a}$$

let $n = \sqrt{a \cdot \sqrt{a \cdot \sqrt{a \dots \infty}}}$

(a>0)

$$n = \sqrt{a \cdot n}$$

$$n = a \cdot n \quad (\text{if } b \text{ is true})$$

$$n = a$$

$$n \sqrt{a \cdot \sqrt{a \cdot \sqrt{a \dots \infty}}} = n \sqrt{a}$$

$$n \sqrt{a \cdot \sqrt{a \cdot \sqrt{a \dots \infty}}} = n \sqrt{a}$$

$$\sqrt{a \cdot \sqrt{a \cdot \sqrt{a \dots \infty}}} = a^{\frac{2^n - 1}{2^n}}$$

$$\sqrt[n]{a \cdot \sqrt[n]{b \cdot \sqrt[n]{a \cdot \sqrt[n]{b \dots \infty}}}}$$

$$n = m \sqrt[n]{a \cdot \sqrt[n]{b \cdot n}}$$

$$n^m = a \cdot n \sqrt[n]{b}$$

$$\frac{n^m}{n} = a \cdot b$$

$$n^{m-1} = a \cdot b$$

$$n = m-1 \sqrt[n]{a \cdot b}$$

$$n = \sqrt{a + \sqrt{a + \sqrt{a + \dots \infty}}} = \frac{\sqrt{4a+1} + 1}{2}$$

$$y = \sqrt{a - \sqrt{a - \sqrt{a - \dots \infty}}} = \frac{\sqrt{4a+1} - 1}{2}$$

$$\begin{aligned} n-y &= 1 \\ ny &= a \\ n+y &= \sqrt{4a+1} \end{aligned}$$

mark: 85.8x7

eg let, $n = \sqrt{86 + \sqrt{86 + \sqrt{86 + \dots \infty}}} = 8$

$$y = \sqrt{86 - \sqrt{86 - \sqrt{86 - \dots \infty}}} = \textcircled{7}$$

let:-

$$n = \sqrt{a+b \sqrt{a+b \sqrt{a+b \dots \infty}}} = \frac{\sqrt{4a+b^2} + b}{2}$$

$$y = \sqrt{a-b \sqrt{a-b \sqrt{a-b \dots \infty}}} = \frac{\sqrt{4a+b^2} - b}{2}$$

$$\begin{aligned} n-y &= b \\ ny &= a \\ n+y &= \sqrt{4a+b^2} \end{aligned}$$

let:-

$$n = \sqrt{a + \sqrt{a - \sqrt{a + \sqrt{a - \dots \infty}}}} = \frac{\sqrt{4a-3} + 1}{2}$$

$$y = \sqrt{a - \sqrt{a + \sqrt{a - \sqrt{a + \dots \infty}}}} = \frac{\sqrt{4a-3} - 1}{2}$$

$$n-y=1$$

if, $n^n = n$, then $n = n \sqrt[n]{n}$

if $n(n+a), (n+2a), (n+3a)$ are
in AP with common diff 'a'.

concept:-

$$n(n+a)(n+2a)(n+3a) + a^4 = \text{product of 4 terms}$$
$$= [n(n+3a) + a^2]^2$$

END